

Aim: Implement working of Stacks (all primitive operations).

1.Implementation using Arrays (Fixed Size/Static).

CODE:

```
#include <iostream>
using namespace std;
class Stack{
    int sp, cap, *a;
public:
    Stack(int s){
        cap = s;
        sp = -1;
        a = new int[cap];
    }
    int pop(){
        if(sp<0){
            cout<<"Stack Underflow";
            return -1;
        }
        return a[sp--];
    }
    int peek(){
        if(sp<0){
            cout<<"Stack is empty!";
            return -1;
        }
        return a[sp];
    }
    void push(int x){
        if(sp>=cap-1){
            cout<<"StackOverflow!";
            return;
        }
        a[++sp] = x;
    }
    bool isEmpty(){
        return sp<0;
    }
    void display(){
        if(isEmpty()){
            cout<<"Stack is Empty";
            return;
        }
        cout<<"Stack Elements: ";
        for(int i = sp; i >= 0; i--){
            cout<<a[i]<<" ";
        }
    }
};
```

```
int main(){
    int n;
    cout<<"Enter the size of stack:";
    cin>>n;
    Stack s(n);
    int ch, val;
    while(true){
        cout<<"\nChoose the operation for stack:";
        cout<<"\n1.Push:";
        cout<<"\n2.Pop:";
        cout<<"\n3.Peek:";
        cout<<"\n4.Display:";
        cout<<"\n5.Exit:";
        cout<<"\nEnter the option:";
        cin>>ch;
        switch(ch){
            case 1:
                cout<<"Enter the value:";
                cin>>val;
                s.push(val);
                break;
            case 2:
                val = s.pop();
                if(val != -1){
                    cout<<val<<" popped from stack";
                }
                break;
            case 3:
                val = s.peek();
                if(val != -1){
                    cout<<"Top element: "<<val<<endl;
                }
                break;
            case 4:
                s.display();
                break;
            case 5:
                cout<<"Exiting...";
                return 0;
            default:
                cout<<"Invalid operation of stack!";
        }
    }
    return 0;
}
```

OUTPUT:

```
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Enter the size of stack:5
Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:1
Enter the value:4

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:1
Enter the value:31

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:1
Enter the value:43

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:1
Enter the value:44

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:1
Enter the value:55
```

```
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Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:4
Stack Elements: 55 44 43 31 4
Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:3
Top element: 55

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:2
55 popped from stack
Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:3
Top element: 44

Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:4
Stack Elements: 44 43 31 4
```

```
Choose the operation for stack:
1.Push:
2.Pop:
3.Peek:
4.Display:
5.Exit:
Enter the option:5
Exiting...
-----
Process exited after 154.9 seconds with return value 0
Press any key to continue . . . |
```

2.Implementation using Linked Lists (Dynamic Size).**CODE:**

```
#include <iostream>
using namespace std;
class Node {
public:
    int data;
    Node *next;
    Node(int val) {
        data = val;
        next = NULL;
    }
};
class Stack {
    Node *sp;
    public:
        Stack() {
            sp = NULL;
        }

    void push(int val) {
        Node *t = new Node(val);
        t->next = sp;
        sp = t;
    }
    int pop() {
        if (sp == NULL) {
            cout << "Stack Underflow!";
            return -1;
        }
        Node *n = sp;
        int t = n->data;
        sp = sp->next;
        delete n;
        return t;
    }
    int peek() {
        if (sp == NULL){
            cout<<"Stack is Empty!";
            return -1;
        }
        return sp->data;
    }
    void display() {
        if (sp == NULL){
            cout<<"Stack is empty!";
            return;
        }
    }
```

```
Node *t = sp;
cout<<"Stack elements: ";
while (t != NULL){
    cout<< t->data << " ";
    t = t->next;
}
cout<<endl;
};

int main() {
    Stack s;
    int ch, val;
    while (true) {
        cout<<"\nChoose the operation for stack:";
        cout<<"\n1.Push";
        cout<<"\n2.Pop";
        cout<<"\n3.Peek";
        cout<<"\n4.Display";
        cout<<"\n5.Exit";
        cout<<"\nEnter the option:";
        cin>>ch;
        switch (ch){
            case 1:
                cout<<"Enter the value:";
                cin >> val;
                s.push(val);
                break;
            case 2:
                val = s.pop();
                if(val != -1){
                    cout<<val<<" popped from stack!";
                }
                break;
            case 3:
                val = s.peek();
                if(val != -1){
                    cout<<"Top element:"<<val<<endl;
                }
                break;
            case 4:
                s.display();
                break;
            case 5:
                cout<<"Exiting...";
                return 0;
            default:
                cout<<"Invalid operation of stack!";
        }
    }
}
```

```
    }  
  }  
  return 0;  
}
```

OUTPUT:

```
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:1  
Enter the value:1  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:1  
Enter the value:2  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:1  
Enter the value:3  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:1  
Enter the value:4  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:1  
Enter the value:5
```

```
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Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:4  
Stack elements: 5 4 3 2 1  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:2  
5 popped from stack!  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:3  
Top element:4  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:4  
Stack elements: 4 3 2 1  
  
Choose the operation for stack:  
1.Push  
2.Pop  
3.Peek  
4.Display  
5.Exit  
Enter the option:5  
Exiting...  
-----  
Process exited after 25.29 seconds  
Press any key to continue . . . |
```